



Android tablet as second screen

I turned my old Huawei MatePad T10 into a real second monitor — and it's awesome

Do you have an old Android tablet collecting dust? Mine is now a **true extended desktop** for my Linux laptop — not a mirror, a real second screen. Windows can be dragged onto it, it has its own resolution and scaling, and it connects over a plain USB cable with ~1 ms latency.

Here's the full build, including every configuration file.

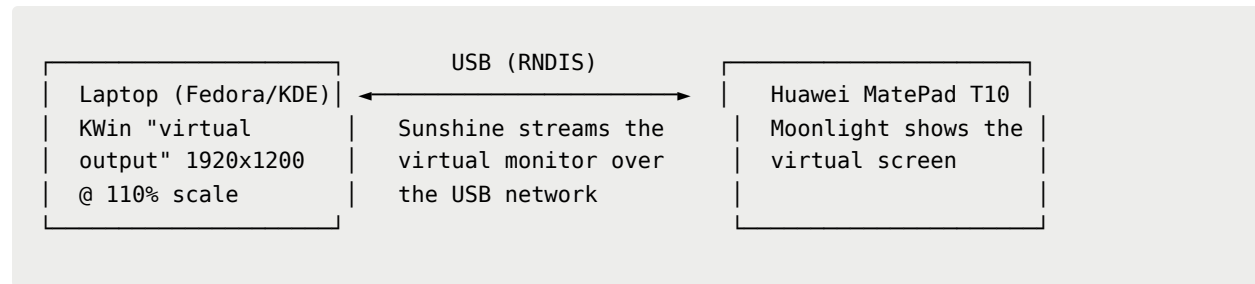
The stack

Component	What it does
Fedora 44 + KDE Plasma 6 (Wayland)	Host OS
krfb-virtualmonitor	Creates a <i>virtual monitor</i> inside KWin — a real extra output that KDE treats like any physical display
Sunshine	Streaming server (same tech as NVIDIA GameStream, but open source)
Moonlight	Android client on the tablet
USB tethering (RNDIS)	Direct USB network link between tablet and PC — no router needed

Why USB? Because it gives a **direct wired network** between the two devices: ~1 ms ping, zero Wi-Fi congestion, and the stream is rock solid.

```
PC:      enp0s20f0u3  192.168.42.226/24
Tablet:  192.168.42.129
Latency: ~1.2 ms
```

How it works



1. `krfb-virtualmonitor` creates a virtual output (1920×1200) that KWin sees as a second screen.
2. It gets positioned **to the left** of the laptop panel and scaled to **110%**.
3. Sunshine captures exactly that output (KWin screencast over PipeWire) and streams it.
4. Moonlight on the tablet displays it — a genuine extended desktop.

Step 1 — The USB network link

1. Plug the tablet into the laptop with a data cable.
2. On the tablet: **Settings** → **More** → **USB tethering**.
3. On the PC, a new `rndis_host` interface appears:

```
$ ip -brief address
enp0s20f0u3    UNKNOWN    192.168.42.226/24    # <- new USB li
nk

$ ping -c 3 192.168.42.129    # the tablet
64 bytes from 192.168.42.129: icmp_seq=1 ttl=64 time=1.29 ms
```

Note: some WiFi-only tablets refuse tethering. If yours does, fall back to putting both devices on the same Wi-Fi and use the laptop's LAN IP instead.

Step 2 — Create the virtual monitor

```
sudo dnf install krfb
```

Then create a 1920×1200 virtual output (it also starts a VNC server on the same screen, which is a handy fallback):

```
krfb-virtualmonitor --resolution 1920x1200 \  
  --name matepad \  
  --password matepad2026 \  
  --scale 1 \  
  --port 5900
```

You'll see a new output appear in **System Settings** → **Display and Monitor**:

Output: 1 Virtual-matepad	enabled	1920x1200 @ 60
Output: 2 eDP-1	enabled	1920x1080

Step 3 — Install & configure Sunshine

```
sudo dnf copr enable lizardbyte/stable  
sudo dnf install Sunshine
```

```
~/.config/sunshine/sunshine.conf:
```

```
capture= kwin          # capture via KWin screencast (Wayland) - required for virtual out  
puts  
output_name= Virtual-matepad  
encoder= software      # libx264 (see "encoder notes" below)  
fps=60  
bitrate=20000
```

Start it and open the web UI at <https://localhost:47990> to set credentials and pair clients:

sunshine

Sunshine log (the important lines):

```
Screencasting output name Virtual-matepad position 0x0 resolution 1920x1200  
Found H.264 encoder: libx264 [software]
```

Step 4 — Tablet client

1. Install **Moonlight Game Streaming** from the Play Store (or sideload from moonlight-stream.org).
2. Add a host manually: 192.168.42.226.
3. Pair with the PIN shown in the Sunshine web UI.
4. Tap the **Desktop** app → the tablet becomes your second monitor.

Step 5 — Position + scale (the annoying part, solved)

KScreen does **not** support gaps between displays, and scaling changes the logical width. So you can't just hardcode positions — here's the fix:

```
# virtual screen on the LEFT at 110% scale  
kscreen-doctor output.Virtual-matepad.position.0,0 output.Virtual-matepad.scale.1.1  
  
# laptop panel must start exactly at the virtual screen's right edge  
# (1920 / 1.1 ≈ 1746, NOT 1920 - a hardcoded 1920 leaves a 174px gap!)  
kscreen-doctor output.eDP-1.position.1746,0
```

Result:

```
Output: 1 Virtual-matepad      Geometry: 0,0      1746x1091  Scale: 1.1
Output: 2 eDP-1               Geometry: 1746,0  1920x1080  Scale: 1
```

Lesson learned: KDE rejects negative coordinates, and it rejects gaps. Align the primary screen to the *computed* edge of the scaled virtual screen.

Step 6 — Automate everything

`~/.local/bin/matepad-on.sh` — idempotent start script:

```
#!/usr/bin/env bash
set -euo pipefail

LOG=/tmp/matepad-screen.log
RESOLUTION=1920x1200
NAME=matepad
PASSWORD=matepad2026
PORT=5900
VMON="Virtual-$NAME"

log() { echo "[$(date '+%H:%M:%S')] $" >> "$LOG"; }

# 1. Start the virtual monitor if it isn't already running
if !pgrep -f '^krfb-virtualmonitor ' >/dev/null; then
    log "starting krfb-virtualmonitor"
    nohup krfb-virtualmonitor --resolution "$RESOLUTION" --name
"$NAME" \
    --password "$PASSWORD" --scale 1 --port "$PORT" >> "$LOG"
2>&1 &
else
    log "krfb-virtualmonitor already running"
fi

# 2. Wait for the output to appear (up to ~10s)
for i in $(seq 1 20); do
```

```

kscreen-doctor -o 2>/dev/null | grep -q "$VMON" && break
sleep 0.5
done

# 3. Position left, scale 110%, align primary to the computed
right edge
kscreen-doctor output.$VMON.position.0,0 output.$VMON.scale.
1.1
sleep 1
VMW=$(kscreen-doctor -o 2>/dev/null | sed 's/\x1b\[([0-9;])*m//
g' | \
    awk -v v="$VMON" '$0 ~ v {f=1} f && /Geometry:/ {split($3,
d,"x"); print d[1]; exit}')
if [ -n "$VMW" ]; then
    kscreen-doctor output.eDP-1.position.${VMW},0
    log "positioned$VMON at 0,0 (left) scale 1.1, eDP-1 at${VM
W},0"
else
    kscreen-doctor output.eDP-1.position.1920,0
    log "positioned$VMON at 0,0 (left) scale 1.1 (width fallback
1920)"
fi

# 4. Start Sunshine if it isn't running
if !pgrep -x sunshine >/dev/null; then
    log "starting sunshine"
    nohup sunshine >> "$LOG" 2>&1 &
else
    log "sunshine already running"
fi

log "done"

```

```
~/.local/bin/matepad-off.sh :
```

```
#!/usr/bin/env bash
LOG=/tmp/matepad-screen.log
log() { echo "[$(date '+%H:%M:%S')] $" >> "$LOG"; }

if pgrep -x sunshine >/dev/null; then
    pkill -x sunshine && log "stopped sunshine"
fi
if pgrep -f '^krfb-virtualmonitor ' >/dev/null; then
    pkill -f '^krfb-virtualmonitor ' && log "stopped krfb-virtualmonitor"
fi
log "done"
```

`~/.config/autostart/matepad-screen.desktop` — so it all comes up at login:

```
[Desktop Entry]
Type=Application
Name=MatePad Second Screen
Comment=Start MatePad virtual monitor and Sunshine at login
Exec=/home/you/.local/bin/matepad-on.sh
Terminal=false
X-KDE-autostart-after=panel
X-KDE-autostart-phase=2
```

Now the daily routine is: **plug in the cable** → **enable tethering** → **open Moonlight** → **tap fedora**. No terminal commands.

Encoder notes (the rabbit hole I saved you from)

I have a hybrid GPU laptop (Intel UHD 630 + NVIDIA GTX 1050). Every hardware encoding path failed:

Path	Result
Intel VAAPI (<code>h264_vaapi</code>)	✗ 2026 iHD driver dropped Coffee Lake encode ("No usable encoding profile")
Sunshine <code>encoder=nvenc</code>	✗ bundled CUDA no longer ships kernels for Pascal GPUs ("no kernel image available")

Path	Result
NVIDIA via <code>libva-nvidia-driver</code> (VAAPI)	✗ encode profiles unavailable
Software <code>libx264</code>	✓ works flawlessly — this is what I use

`sudo dnf install libva-nvidia-driver` doesn't hurt to have installed, but on this hardware software encoding over the USB link is crisp.

Troubleshooting cheatsheet

Symptom	Fix
Gaps between displays are not supported	Align <code>eDP-1</code> to the computed edge of the scaled virtual screen (<code>1746,0</code>), not <code>1920,0</code>
KWin screencast permission error (Error 503)	Set <code>KWIN_WAYLAND_NO_PERMISSION_CHECKS=1</code> in <code>~/.config/environment.d/kwin.conf</code> , re-login
Moonlight can't find the host	Add it manually by IP (<code>192.168.42.226</code>)
Stream is choppy	Lower <code>bitrate</code> / <code>fps</code> in <code>sunshine.conf</code>
Virtual output missing	Re-run <code>matepad-on.sh</code> — it recreates it and re-applies position + scale

Final result

- Tablet is a **real extended monitor on the left**, at its native 1920×1200 scaled to 110%
- Direct USB network: ~1 ms latency
- Fully automated at login, `matepad-on.sh` / `matepad-off.sh` to toggle
- Total cost: the USB cable and a bit of config time

If you've got an old tablet lying around, this is one of the most satisfying DIY upgrades — a free second monitor with none of the cloud-screen subscription nonsense.